

FEBRUARY

22

Tuesday

UNIT-1 MATRIX AND DETERMINATES

Week 49
53-54

Q1 Solve for x :

$$\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-2 & 2x-9 & 3x-16 \\ x-0 & 2x-27 & 3x-64 \end{vmatrix} = 0$$

Q2 Find the inverse of matrix

$$\begin{vmatrix} 3 & -10 & -1 \\ -2 & 0 & 2 \\ 2 & -4 & -2 \end{vmatrix}$$

Q3 Solve these simultaneous equations
Cramer's rule.

$$x + y + 2z = 4$$

$$2x - y - z = 1 \quad \text{and}$$

$$3x - 2y - z = 3$$

Q4 Find the inverse of matrix

$$\begin{vmatrix} 1 & 2 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{vmatrix}$$

Q5 Short Notes

(a) Cramer's Rule

(b) Transpose of matrix

(c) Symmetric and Skew Symmetric matrix

How we live is important, not how long.

February

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 • •

Q6 If $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & 0 & 2 \\ 1 & -3 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 4 & 5 & 6 \\ -1 & 0 & 1 \\ 2 & 1 & 2 \end{pmatrix}$

$C = \begin{pmatrix} +1 & -2 & 1 \\ -1 & 2 & 3 \\ -1 & -2 & 2 \end{pmatrix}$ find $A - 2B + 3C$

Q7 Using properties of determinants, prove the following.

$$\begin{vmatrix} 3a & -a+b & -a+c \\ a-b & 3b & c-b \\ a-c & b-c & 3c \end{vmatrix} = 3(a+b+c)(ab+bc+ca)$$

Q8 Using properties of determinants, prove the following.

$$\begin{vmatrix} a & b & c \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{vmatrix} = a^3 + b^3 + c^3 - 3abc$$

Q9 Solve the following system of equations by matrix method

$$x - y + z = 4$$

$$x - 2y - 2z = 9$$

$$2x + y + 3z = 1$$

Human nature is greedy of novelty.

FEBRUARY

24

Thursday

⑨ ⑩ Write a short notes

① Diagonal matrix

② Scalar matrix

③ Rectangular matrix